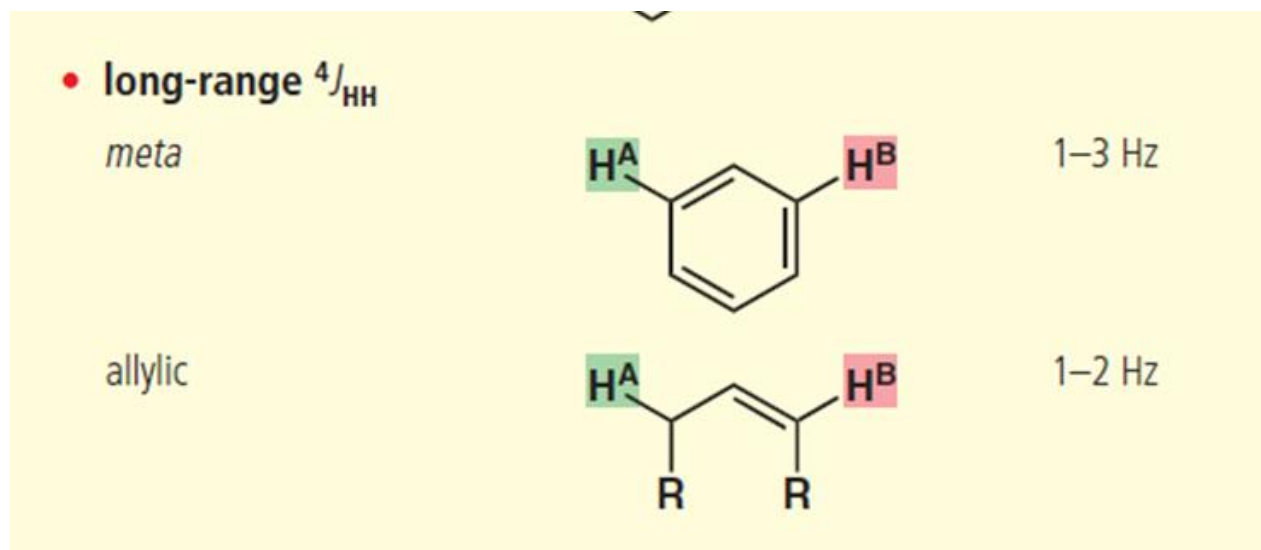
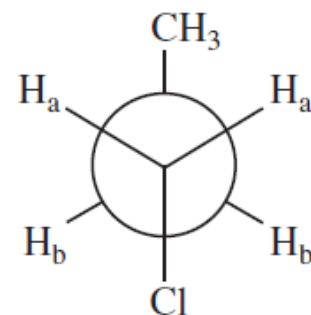
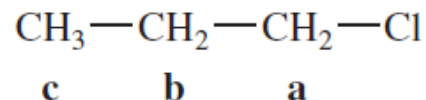


## Long-range coupling

When the through-bond distance gets longer than three bonds, coupling is not usually seen. However, it is seen in some special cases, the most important being *meta* coupling in aromatic rings and allylic coupling in alkenes. In both, the orbitals between the two hydrogen atoms can line up in a zig-zag fashion to maximize interaction. This arrangement looks rather like a letter 'W' and this sort of coupling is called W-coupling. Even with this advantage, values of  $^4J_{\text{HH}}$  are usually small, about 1–3 Hz.

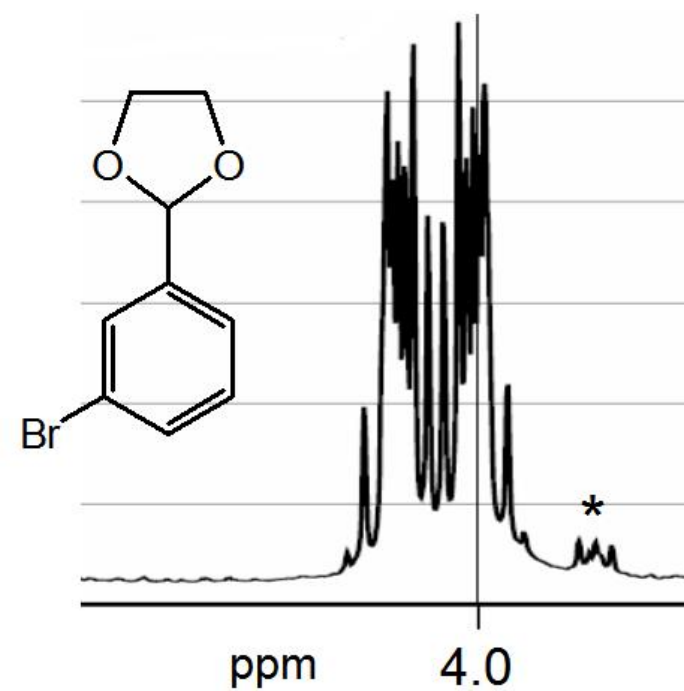
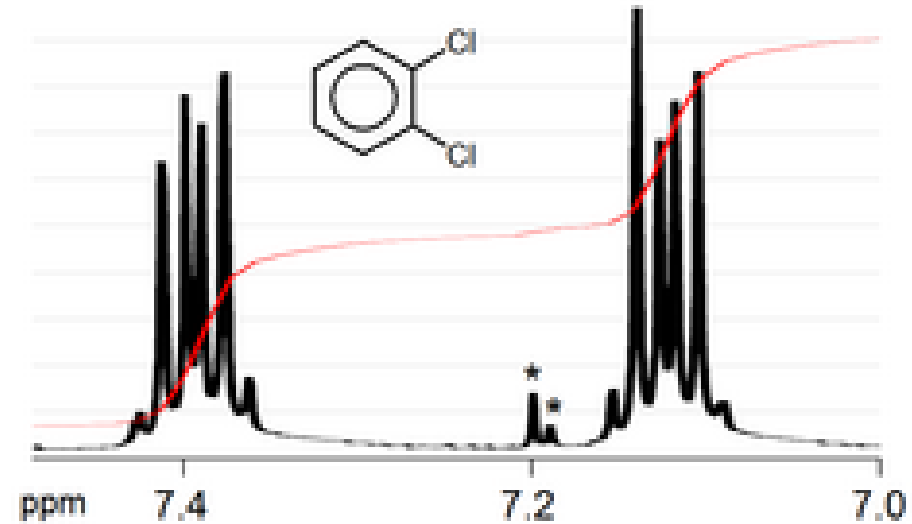
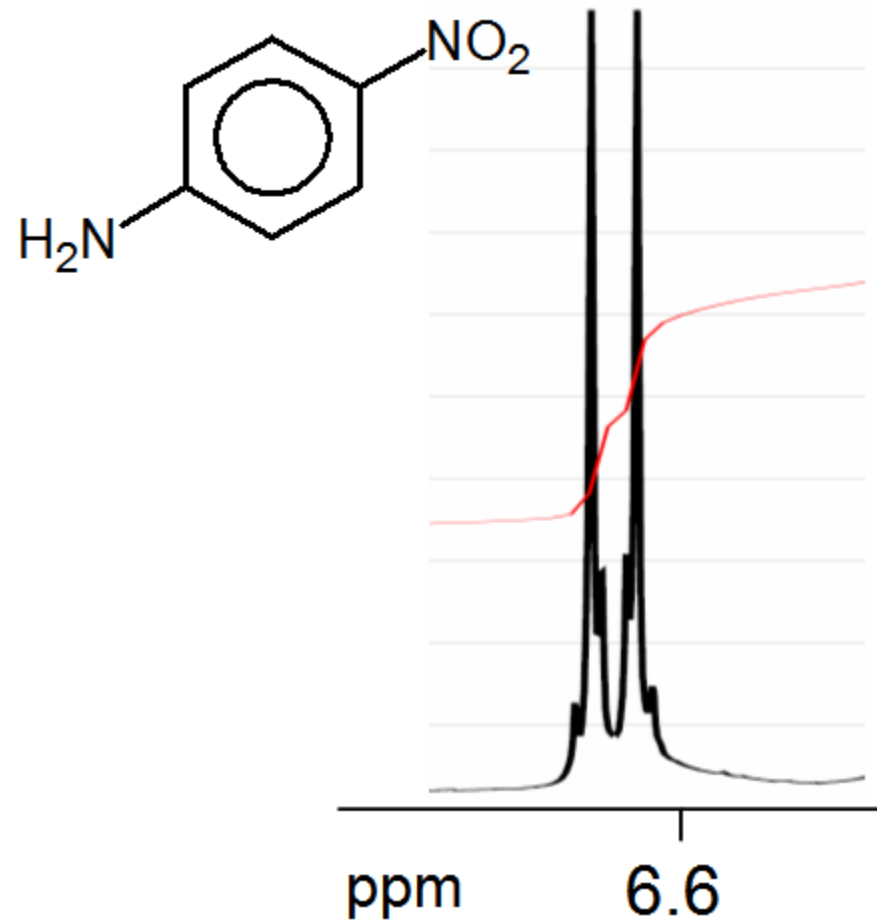


1-Chloropropane



If conformation  
is locked  
(no rotation)

Now consider 1-chloropropane. The hydrogens within a group (those on C1, C2, and C3) are isochronous, but each group is on a different carbon, and as a result, each *group* of hydrogens has a different chemical shift. The hydrogens in each group experience identical *average* magnetic environments, mainly because of free rotation, and are magnetically equivalent. Furthermore, also because of rotation, the hydrogens in each group are equally coupled to the hydrogens in the other groups. If we consider the two hydrogens on C2,  $H_b$  and  $H_b'$  and pick any other hydrogen on either C1 or C3, both  $H_b$  and  $H_b'$  will have the same coupling constant to the designated hydrogen. Without free rotation (see the preceding illustration) there would be no magnetic equivalence. Because of the fixed unequal dihedral angles ( $H_a|Cl|Cl|C_b$  versus  $H_a|Cl|Cl|H_b'$ ),  $J_{ab}$  and  $J_{ab'}$  would not be the same. Free rotation can be slowed or stopped by lowering the temperature, in which case  $H_b$  and  $H_b'$  would become magnetically inequivalent. This type of magnetic inequivalence is often seen in 1,2-disubstituted ethane groups in which the substituents have sufficient steric bulk to hinder free rotation around the C-C axis enough that it becomes slow on the NMR time-scale.



**100.** Reaction of styrene ( $\text{PhCH}=\text{CH}_2$ ) with HBr gives a mixture of regioisomers **A** (major) and **B** (minor). The  $^1\text{H}$  NMR spectrum of the mixture shows four signals, amongst others, at  $\delta$  5.17, 3.53, 3.15 and 2.00 ppm with relative integration of 2:1:1:6, respectively. The molar ratio of **A** and **B** is

1. An organic compound ( $C_7H_{12}O_2$ ) exhibited the following data in the  $^1H$  NMR spectrum.

$\delta$  7.10 (1H, dt,  $J = 16$  Hz and  $7.2$  Hz), 5.90 (1H, dt,  $J = 16$  Hz and  $2$  Hz), 4.1 (2H, q,  $J = 7.2$  Hz), 2.10 (2H, m), 1.25 (3H, t,  $J = 7.2$  Hz), 0.90 (3H, t,  $J = 7.2$  Hz) ppm. (CSIR NET)

Q2. Determine the correct structure of a compound having molecular formula  $C_6H_{11}BrO_2$ , based on the following characteristic spectral data:

**IR:**  $1736\text{ cm}^{-1}$

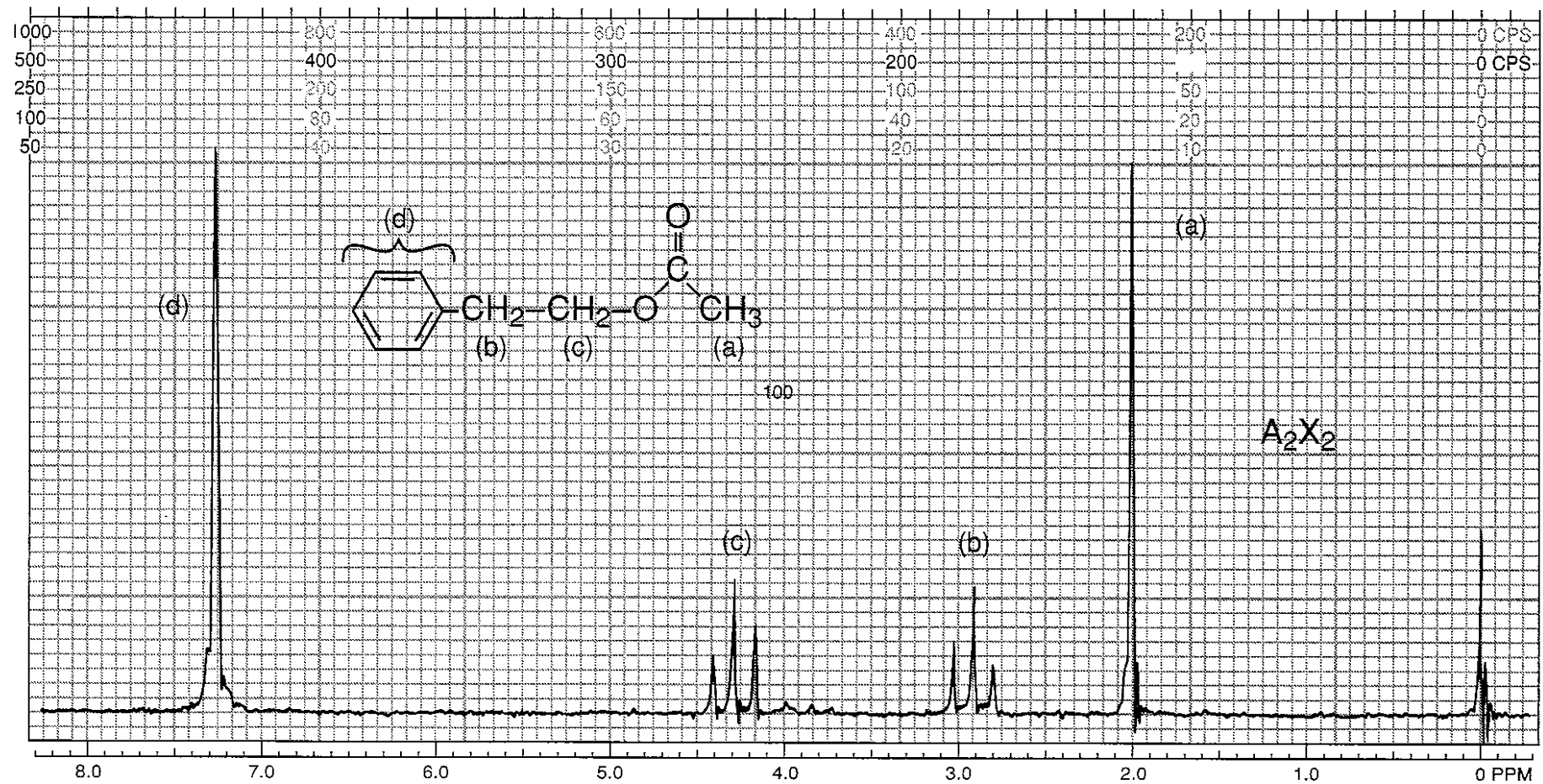
**$^1H$  NMR ( $\delta$ , ppm):** 3.59 (s, 3H), 3.32 (t, 2H), 2.25 (t, 2H), 1.85–1.75 (m, 2H), 1.73–1.62 (m, 2H)

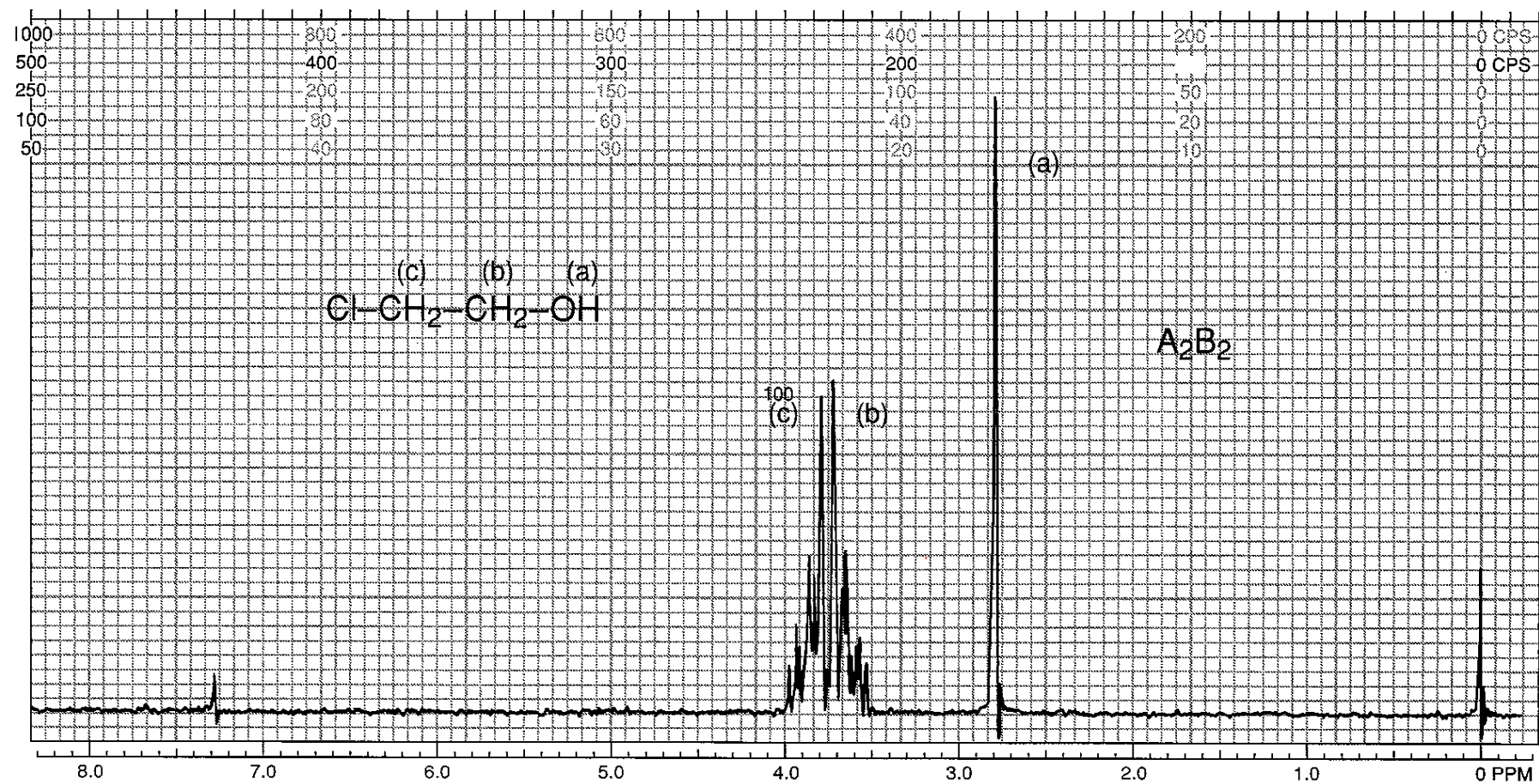
**$^{13}C$  NMR ( $\delta$ , ppm):** 174.0, 51.0, 32.9, 32.8, 31.0, 23.0

(CSIR NET)

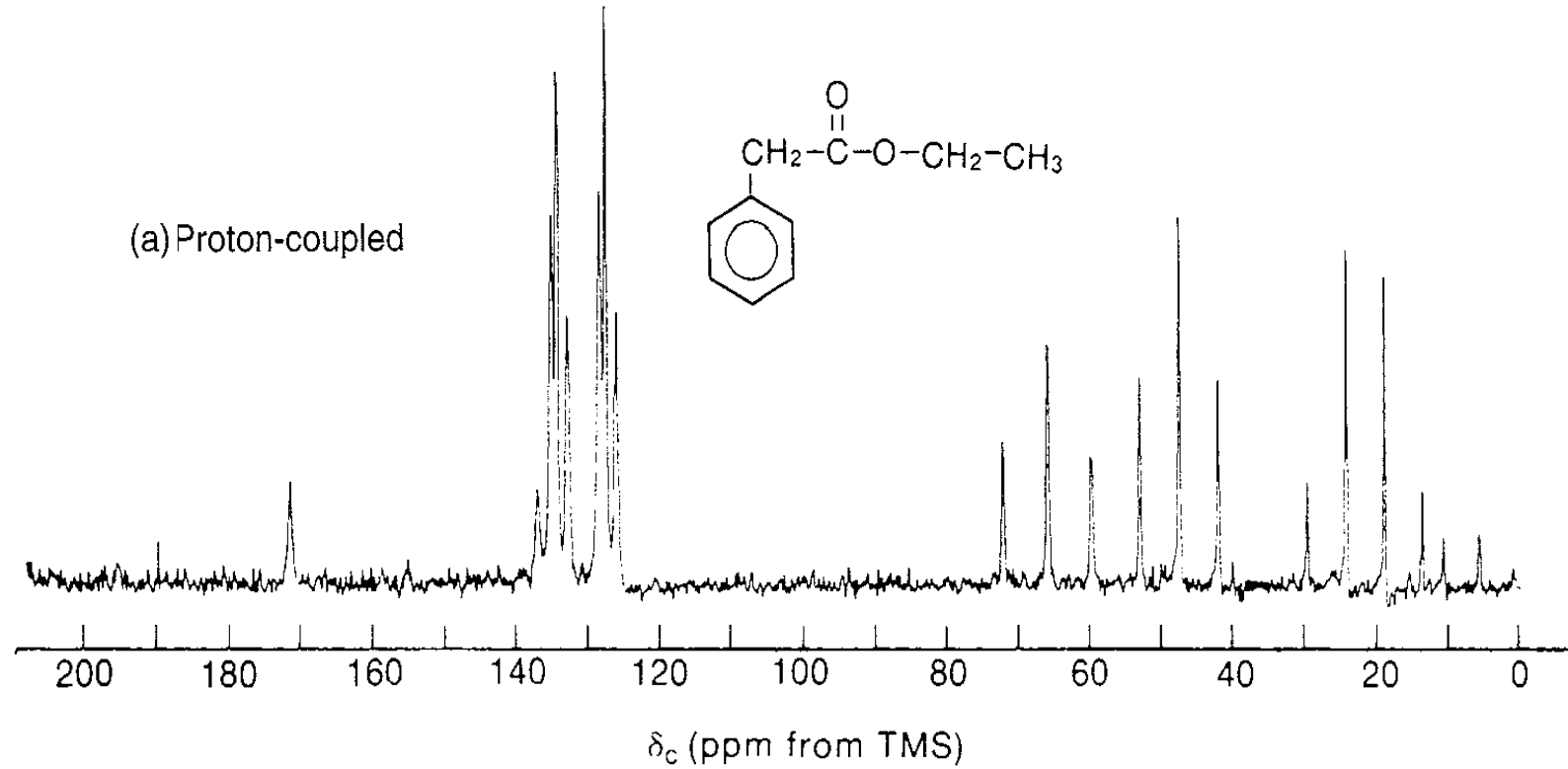
Q. Isomeric compounds **A** and **B** have the composition  $C_{11}H_{12}O_4$ . Deduce structures for **A** and **B** from the spectral data (in the proton NMR, ortho and meta couplings are visible). **Compound A**: IR (KBr): 1720 (s), 1240 (s)  $cm^{-1}$ ,  $^1H$  NMR ( $CDCl_3$ ):  $\delta$  2.46 (s, 3H), 3.94 (s, 6H), 8.05 (d,  $J$  2 Hz, 2H), 8.49 (t,  $J$  2 Hz, 1H); **Compound B**: IR (Nujol<sup>®</sup>): 1720 (s), 1245 (s)  $cm^{-1}$ ,  $^1H$  NMR ( $CDCl_3$ ):  $\delta$  2.63 (s, 3H), 3.91 (s, 6H), 7.28 (d,  $J$  8 Hz, 1H), 8.00 (dd,  $J$  8 Hz, 2 Hz; 1H), 8.52 (d,  $J$  2 Hz, 1H). Assign all protons to the appropriate peaks.

$$DBE: C - (H/2) - (X/2) + (N/2) + 1$$





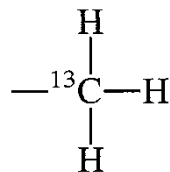
## Proton Coupled Spectra



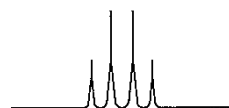
- ✓ The signal intensity is typically low in  $^{13}\text{C}$  NMR
- ✓ Coupling to protons make the intensity of a peak even lower

## Proton Coupled $^{13}\text{C}$ spectra: Splitting of $^{13}\text{C}$ Signals

3 protons

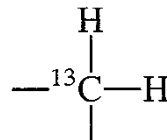


$$\begin{aligned} n + 1 &= 3 + 1 \\ &= 4 \end{aligned}$$

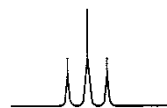


Methyl  
carbon

2 protons

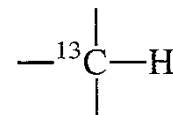


$$n + 1 = 3$$

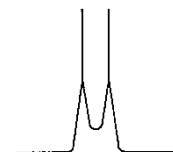


Methylene  
carbon

1 proton

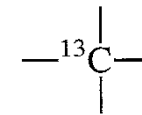


$$n + 1 = 2$$

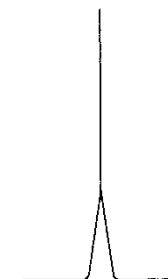


Methine  
carbon

0 protons



$$n + 1 = 1$$

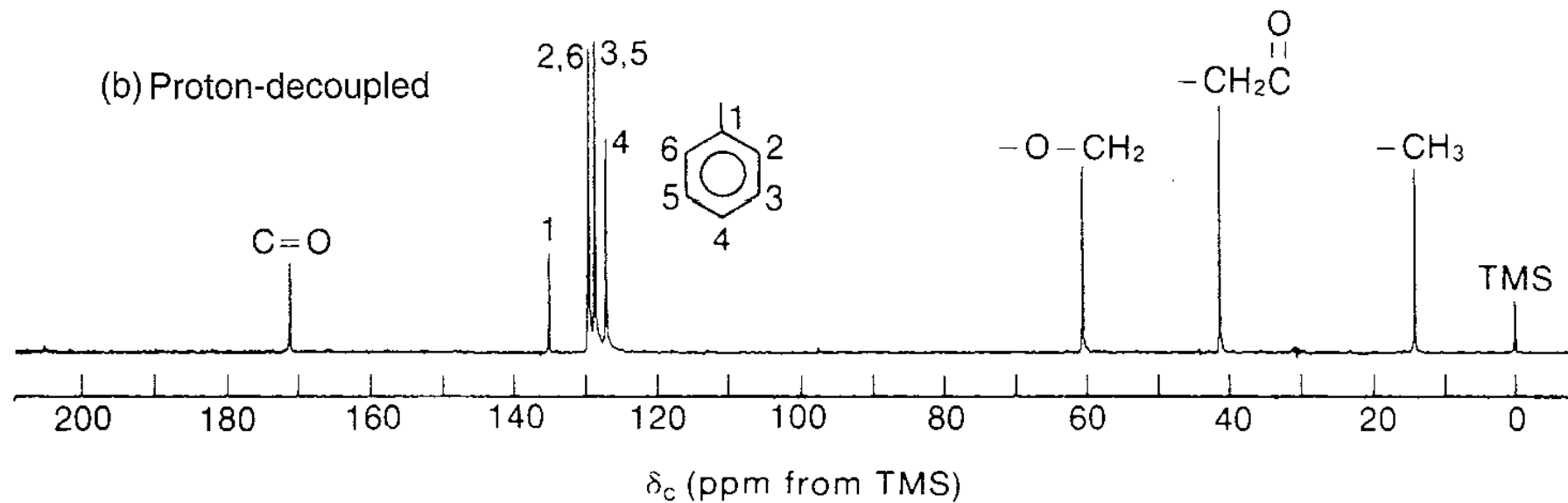


Quaternary  
carbon

✓ The coupling constants ( $^1J$ ) are typically 100-200 Hz

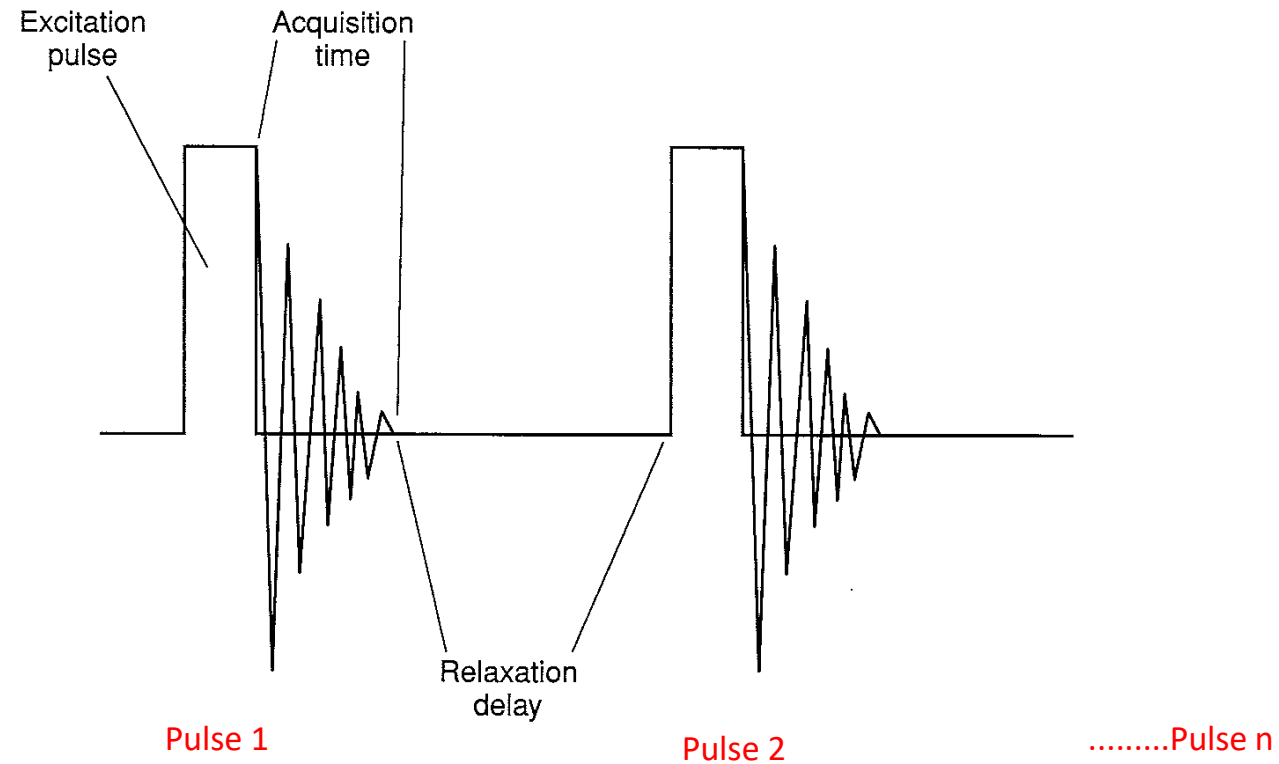
## Proton Decoupled Spectra

- Modern NMR spectroscopy does a special trick
- It removes all the couplings from the protons



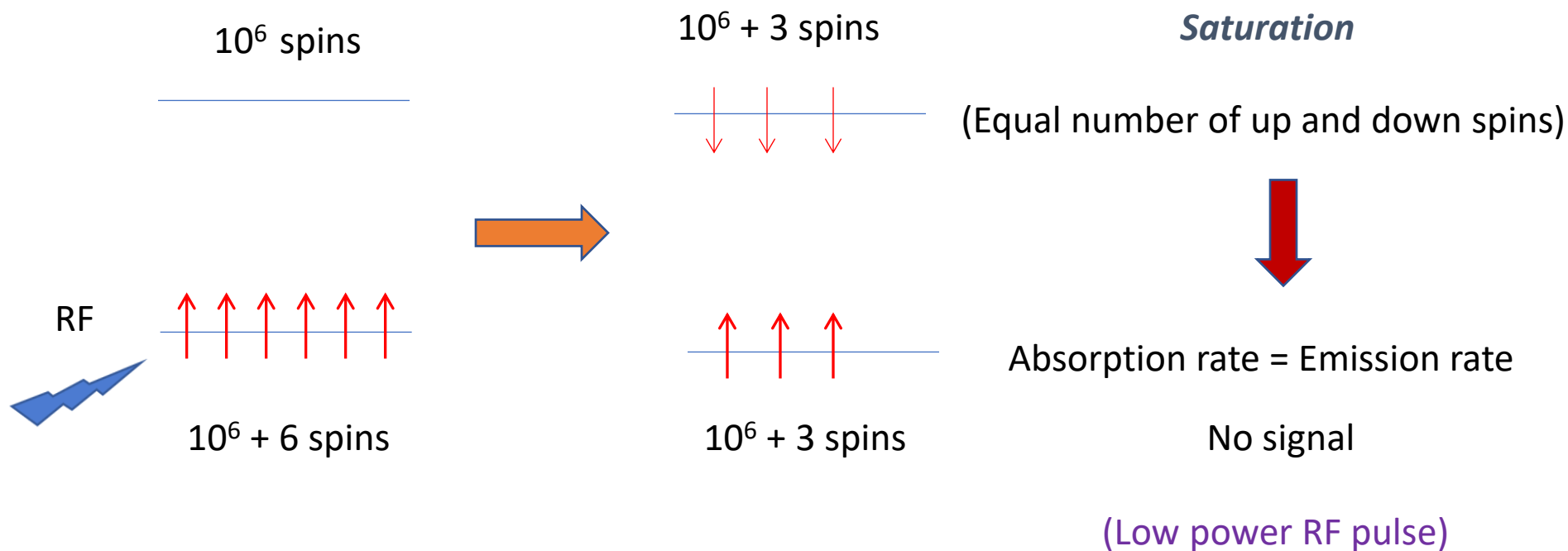
## *How does the decoupling technique work?*

**$^{13}\text{C}$  channel:**



**$^1\text{H}$  channel:** The protons are irradiated continuously with low power RF pulse

## Signal Saturation in NMR



$$\Delta E = h\nu$$

UV-Vis: 400 nm;  $\Delta E = 72$  kcal/mol

IR: 1700  $\text{cm}^{-1}$ ;  $\Delta E = 4.87$  kcal/mol

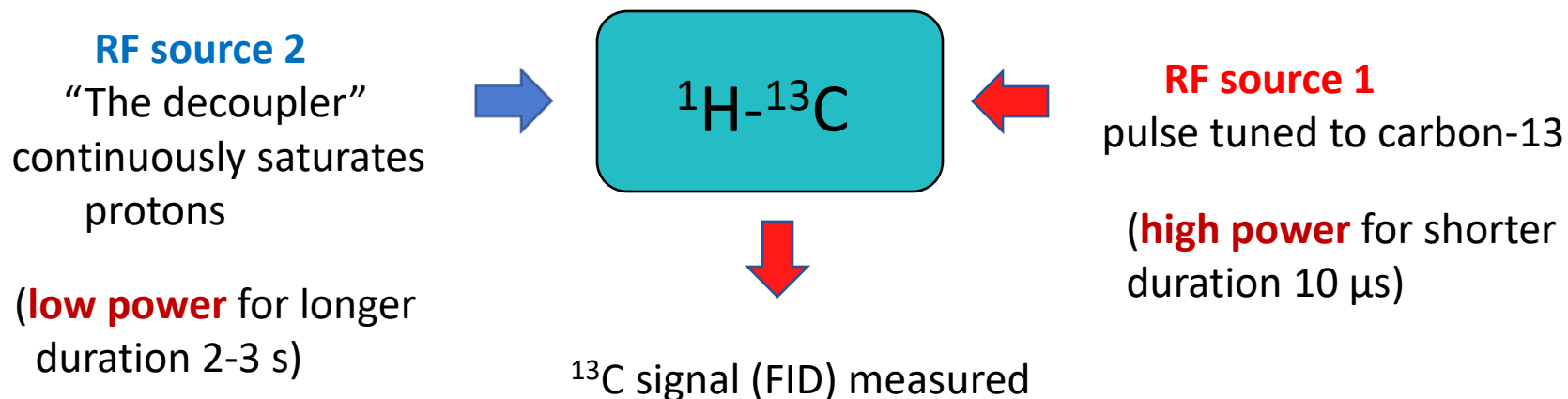
NMR: 500 MHz;  $\Delta E = 0.0477$  cal/mol

$$N_1/N_0 = e^{-\Delta E/kT}$$

*Boltzmann  
Distribution*

## Proton Decoupled Spectra

- All peaks appear as singlets
- While recording a  $^{13}\text{C}$  spectrum, all the protons are simultaneously irradiated in the molecule with a broad spectrum of frequencies in the proper range
- Irradiation causes the protons to become saturated and they undergo rapid upward and downward transitions among all their possible spin states



## ***Nuclear Overhauser Enhancement (NOE)***

- In proton decoupled  $^{13}\text{C}$  spectrum, intensities of the many carbon resonances increase significantly compared to that observed in proton-coupled spectra
- Enhancement increases with the number of hydrogens connected to the carbon nucleus. The degree of enhancement is known as NOE

$$\text{NOE}_{\text{max}} = \frac{1}{2} \left( \frac{\gamma_{\text{irr}}}{\gamma_{\text{obs}}} \right)$$

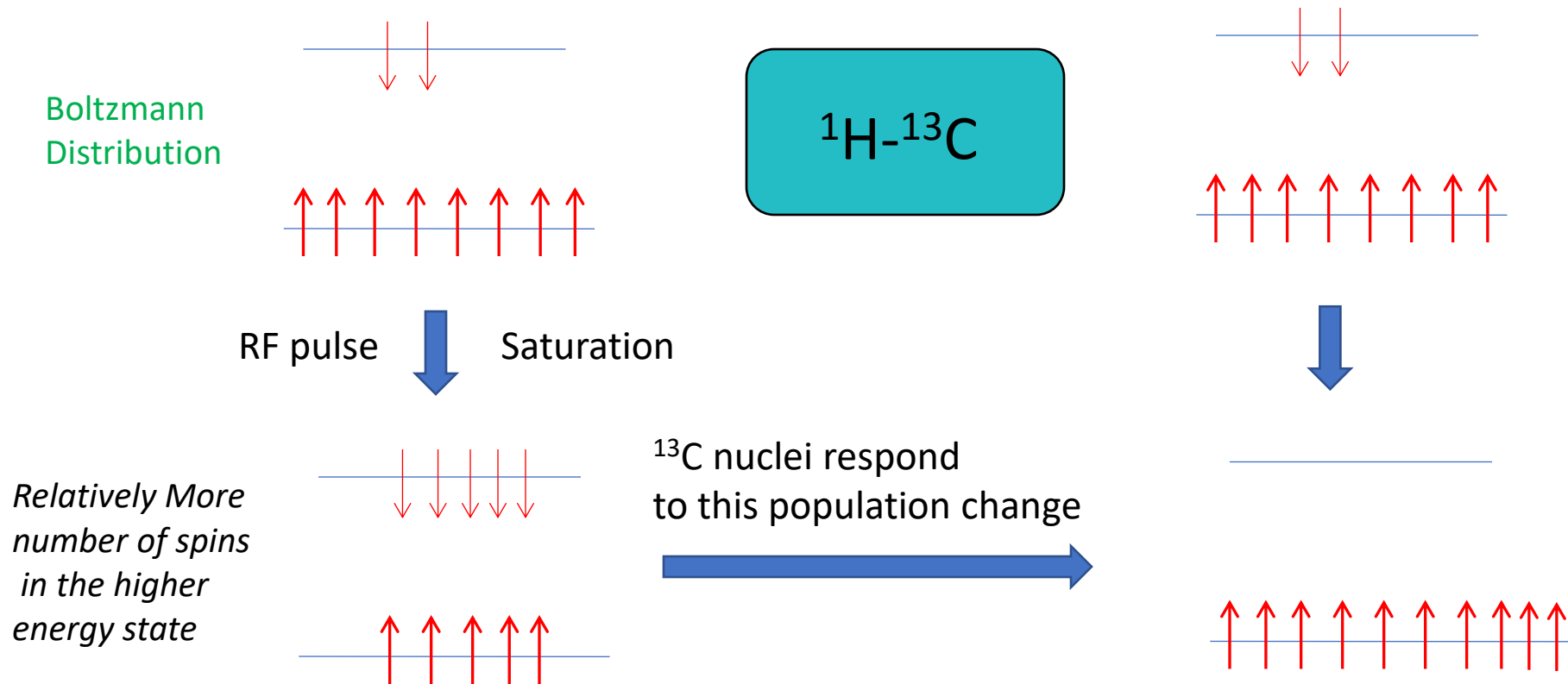
$$\text{NOE}_{\text{max}} = \frac{1}{2} \left( \frac{267.5}{67.28} \right) = 1.988$$

- Theoretically  $^{13}\text{C}$  signals can be enhanced up to 200%

- Proton decoupling not only removes the coupling
- It also increases the signal intensity of the carbons

### ***Nuclear Overhauser Enhancement (NOE)***

- Alters equilibrium population difference between spin states
- Signal enhancement by NOE is a type of *cross-polarization*



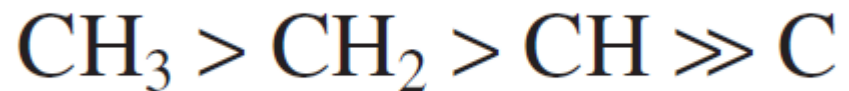
➤ Intensity of NMR signal is proportional to the population difference

*NOE operates through space*

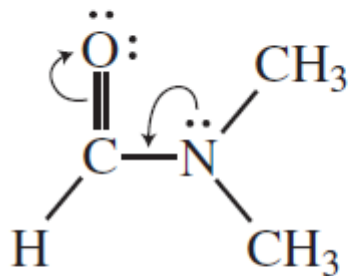
$$\text{NOE} = f\left(\frac{1}{r^3}\right)$$

- In proton decoupled  $^{13}\text{C}$  spectrum, intensities of the many carbon resonances increase significantly compared to that observed in proton-coupled spectra
- Enhancement increases with the number of hydrogens connected to the carbon nucleus

$$\text{NOE}_{\text{max}} = \frac{1}{2} \left( \frac{\gamma_{\text{irr}}}{\gamma_{\text{obs}}} \right)$$



*anti*, 31.1 ppm



*syn*, 36.2 ppm

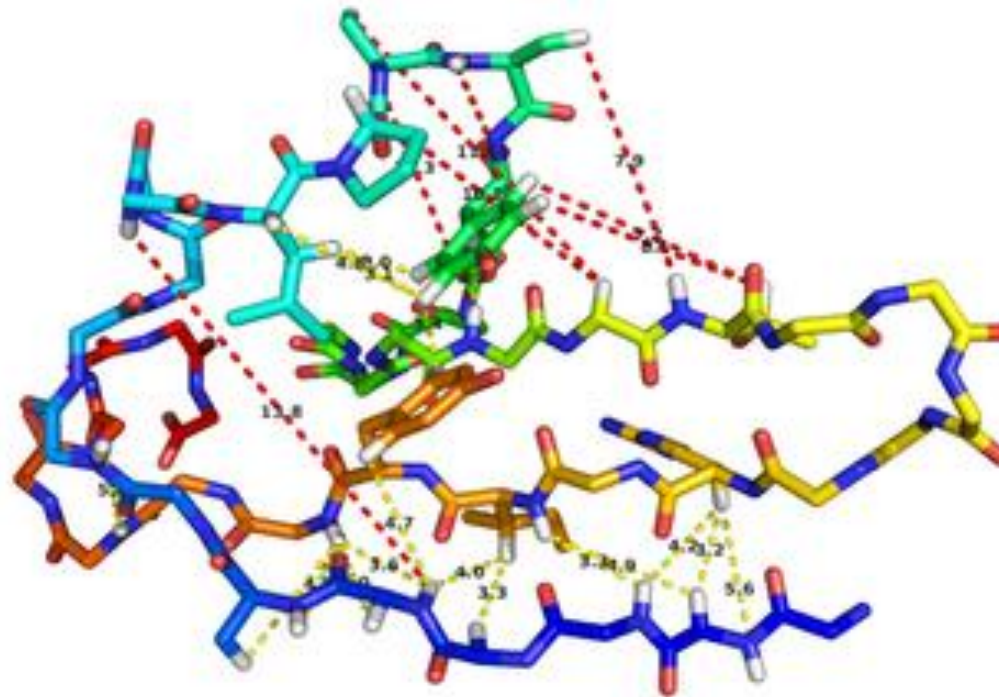
Dimethylformamide

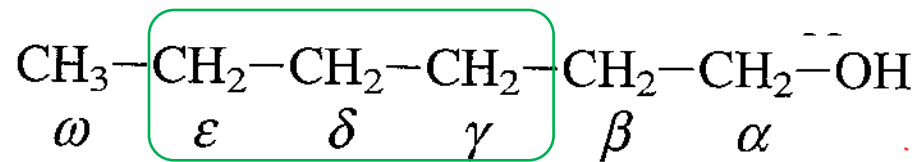
In advanced work, NOEs are sometimes used to verify peak assignments. Irradiation of a selected hydrogen or group of hydrogens leads to a greater enhancement in the signal of the closer of the two carbon atoms being considered. In dimethylformamide, for instance, the two methyl groups are nonequivalent, showing two peaks at 31.1 and 36.2 ppm, because free rotation is restricted about the C-N bond due to resonance interaction between the lone pair on nitrogen and the p bond of the carbonyl group. Irradiation of the aldehyde hydrogen leads to a larger NOE for the carbon of the *syn* methyl group (36.2 ppm) than for that of the *anti* methyl group (31.1 ppm), allowing the peaks to be assigned. The *syn* methyl group is closer to the aldehyde hydrogen.

- Determination of the three-dimensional structures of biological macromolecules in solution
- The inter-atomic distances derived from the observed NOE can often help to confirm the three-dimensional structure of a molecule



Kurt Wüthrich  
(ETH Zurich/Scripps)  
Nobel prize 2002





$^1\text{H}$  NMR      0.90                      1.32                      1.56      3.62

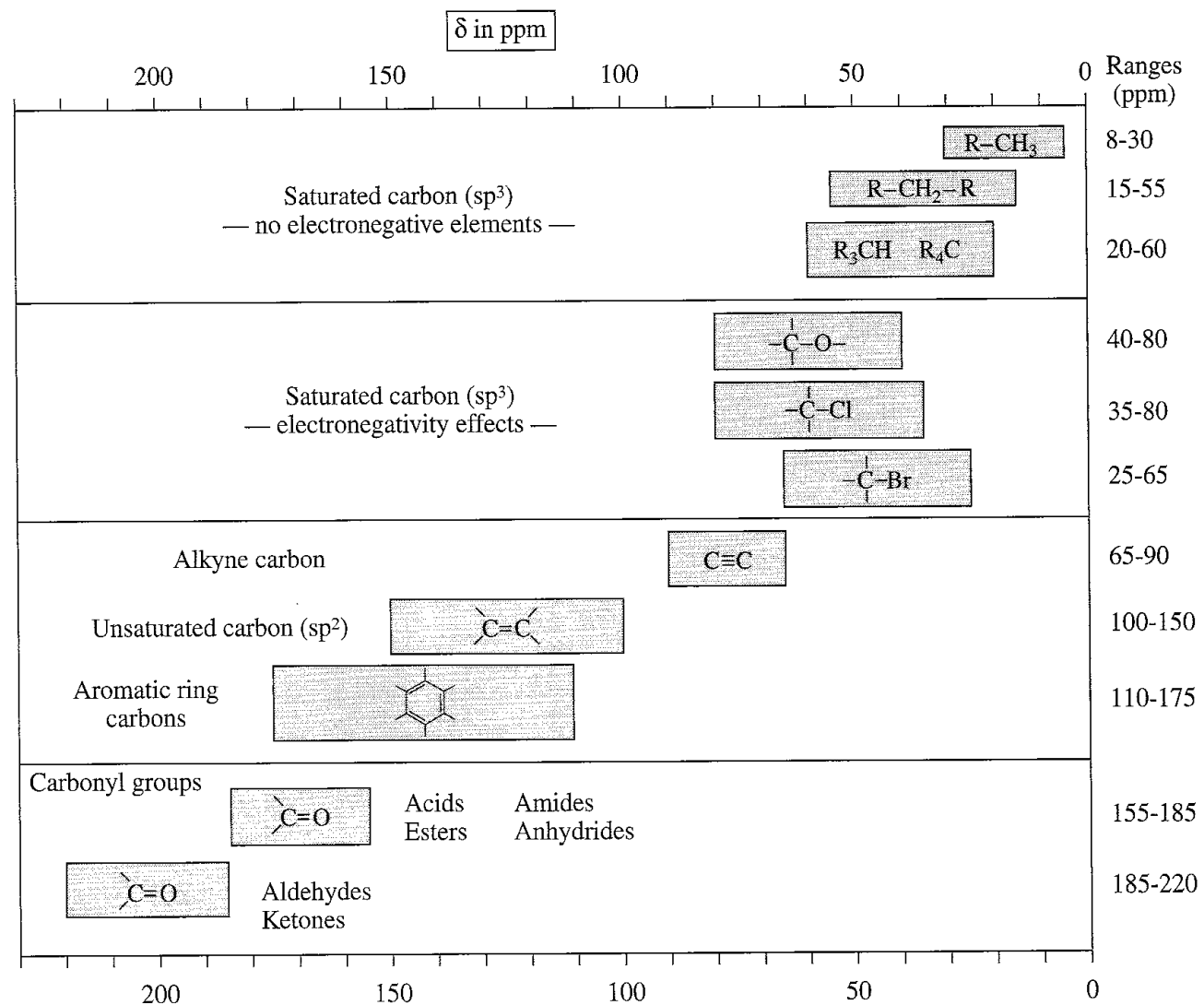
$^{13}\text{C}$  NMR      14.2      22.8      32.0      25.8      32.8      61.9

- Chemical shifts appear over a larger range (0-220 ppm)
- For protons, generally 0-12 ppm

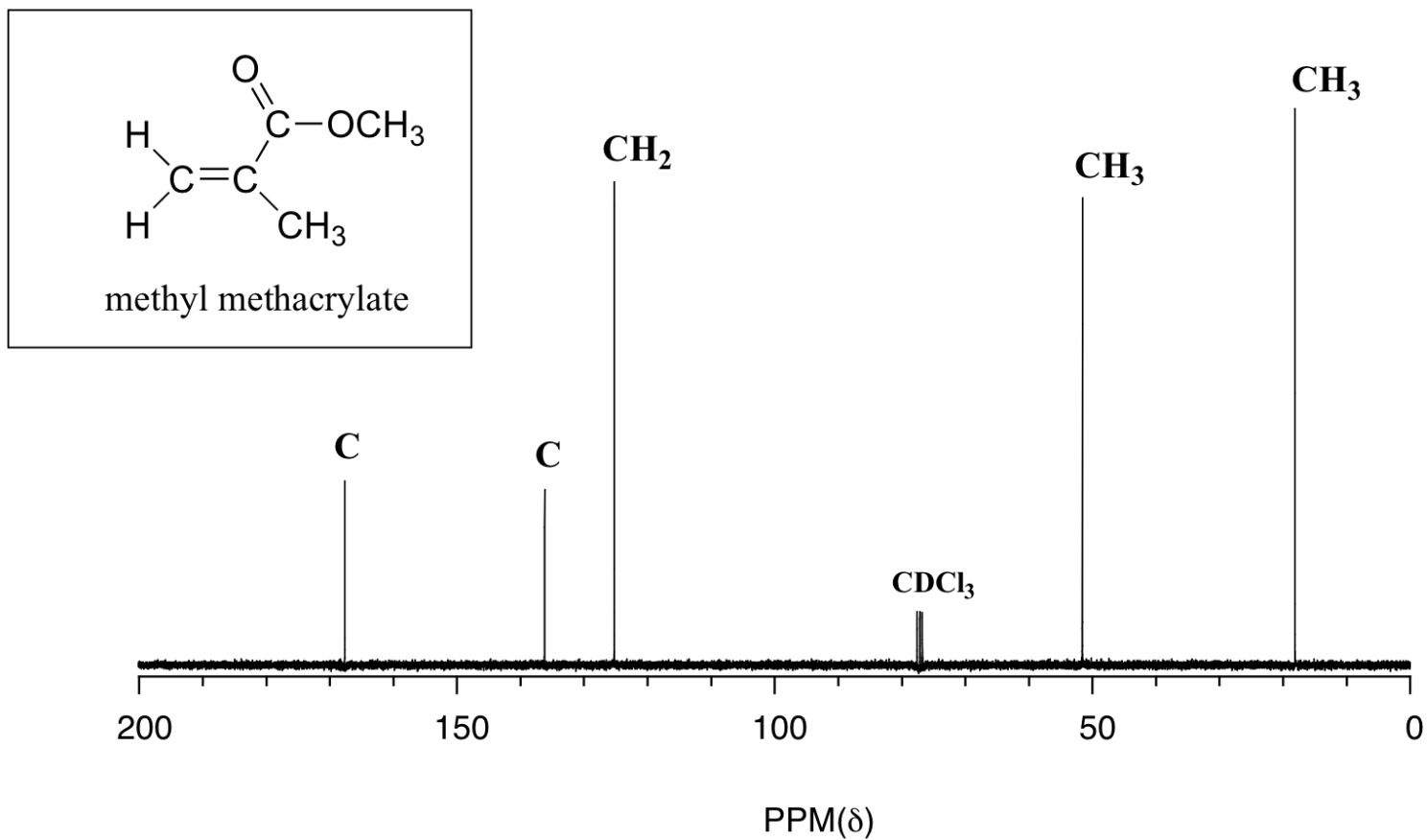


Nearly all non-equivalent carbons in a molecule give separate peaks in  $^{13}\text{C}$  NMR spectra

# <sup>13</sup>C Chemical Shifts



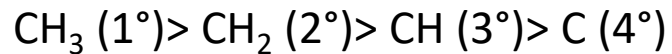
## Problems in Integration of $^{13}\text{C}$ Spectra



## Problems in Integration of $^{13}\text{C}$ Spectra

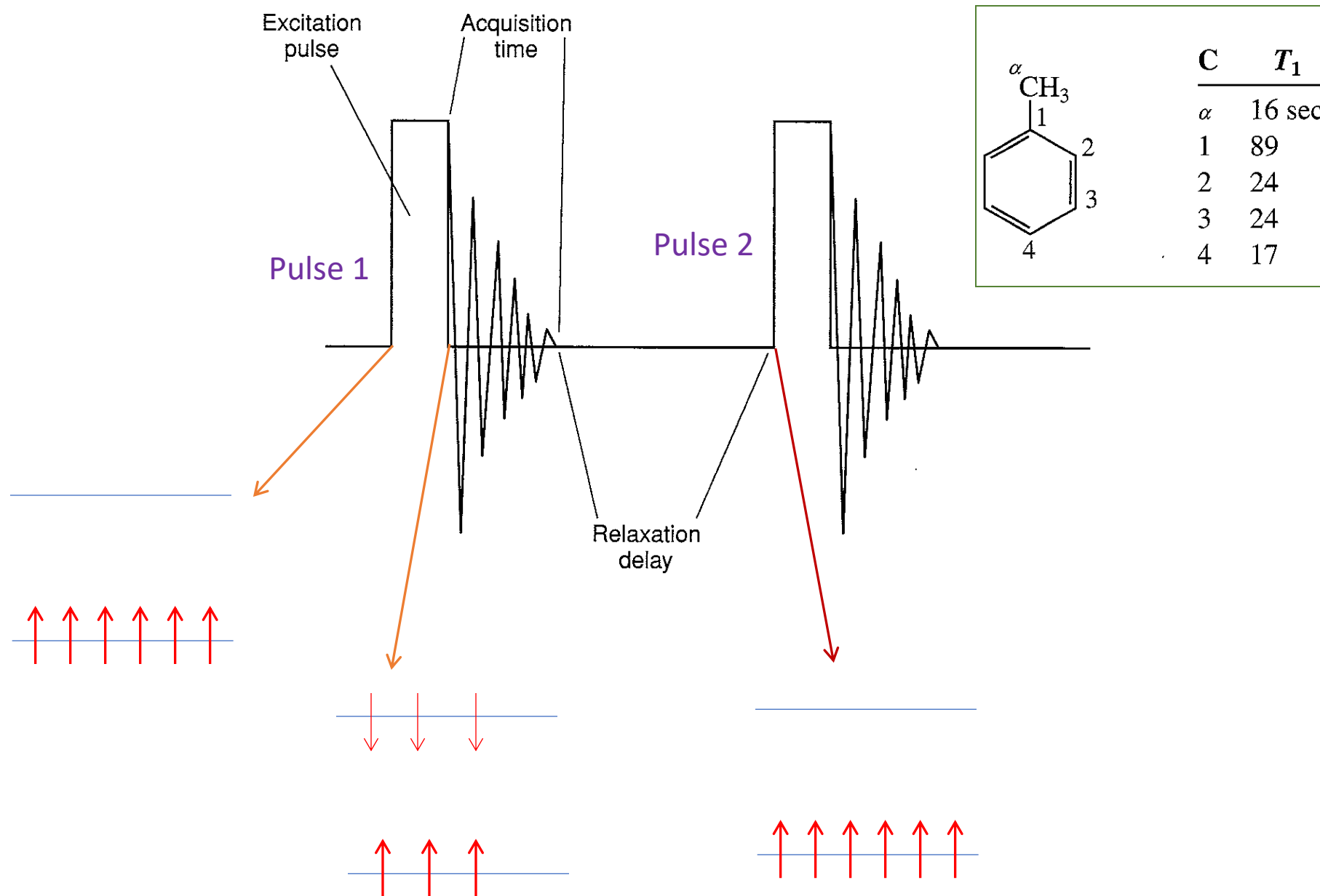
*There are mainly two reasons:*

1. **Nuclear Overhauser Enhancement** is not same for every carbon



The intensity of the carbons with higher number of hydrogens will be more

2. **Variable relaxation time**: For  $^{13}\text{C}$ , some nuclei relax back within seconds but other may require even minutes



- $^{13}\text{C}$  integrals are unreliable → Unlike  $^1\text{H}$  NMR, peak areas should not be used routinely for quantification.
- NOE enhancement distorts intensities → Proton decoupling increases signal unevenly depending on nearby hydrogens.
- Typical intensity trend (approximate):
- $\text{CH}_3 > \text{CH}_2 > \text{CH} > \text{quaternary (weakest)}$
- Relaxation times vary widely ( $T_1$  issue):
- Some carbons relax quickly, others very slowly (seconds to minutes).
- Incomplete relaxation leads to wrong intensities:
- Carbons with long relaxation times give weaker signals → incorrect integrals.
- Pulse timing limitations:
- Short delays between scans prevent full relaxation of all carbons.
- Accurate integration requires special methods:
- Longer delays and acquisition times are needed but are rarely used due to time constraints.